C18 Rate Equations – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Rate Equations and Orders **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	General rate equation				rate = $k[A]^m[B]^n$
2	Order with respect to a reactant				Power of its concentration in the rate equation
3	Overall order				Sum of the individual orders
4	Only factor that changes *k*				Temperature
5	Effect on rate of changing a zero-order concentration				None
6	Order if doubling concentration doubles rate				First
7	Order if doubling concentration quadruples rate				Second
8	Rate-determining step				Slowest step in the mechanism
9	Species that appear in the rate equation				Those in or before the rate-determining step
10	Units of rate				mol dm^{-3} s^{-1}

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C18 Rate Equations – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Units of k and Arrhenius **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Units of *k*: zero order				mol dm^{-3} s^{-1}
2	Units of *k*: first order				s^{-1}
3	Units of *k*: second order				mol^{-1} dm^{3} s^{-1}
4	Units of *k*: third order				mol^{-2} dm^{6} s^{-1}
5	Arrhenius equation				$k = Ae^{-Ea/RT}$
6	Name of *A* in the Arrhenius equation				Pre-exponential (frequency) factor
7	Log form of the Arrhenius equation				$\ln k = -E_a/RT + \ln A$
8	Gradient of ln *k* against 1/*T*				$-E_a/R$
9	y-intercept of ln *k* against 1/*T*				$\ln A$
10	Units of *T* in the Arrhenius equation				K

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C18 Rate Equations – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: Graphs **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Concentration–time graph: zero order				Straight line, negative gradient
2	Concentration–time graph: first order				Curve with constant half-life
3	Rate–concentration graph: zero order				Horizontal line
4	Rate–concentration graph: first order				Straight line through origin
5	Rate–concentration graph: second order				Upward curve through origin
6	Half-life				Time for concentration to halve
7	Half-life of a first-order reaction				Constant
8	Initial rate from a concentration–time graph				Gradient of tangent at $t = 0$
9	Initial rate in a clock reaction is proportional to				$1/t$
10	Value of R				$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.